IBM Fusion

OpenShift Virtualization Lab Guide

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IBM Fusion: TechZone Quick Start

Introduction

IBM Technology Zone (TechZone) is the single destination for our go-to-market teams and IBM business partners ecosystem to access on-demand and live environments to learn, build, show, and share the value of IBM solutions. Additionally, they can extend our certified base images and further customize them for test, education, demonstration, and pilot activities.

This lab guide is a Quick Start for all Fusion labs on TechZone. The lab guide will explain how to find Fusion content, as well as how to reserve and use the Fusion related lab instances. Lab exercises and specific preparation steps are detailed in the respective lab guides by topic of interest.

This guide can be used to bring a TechZone instance to a general starting point. From this starting point you may continue to the relevant lab guide document or, if you wish, create your own specific configuration for a user defined test or demo.

All TechZone instances for Fusion are based on configurable OpenShift clusters with a virtual machine (VM) node infrastructure deployed on OpenShift Virtualization (OCP-Virt) technology. The underlying VMs have nested virtualization enabled so that they can host a second level VM for demonstrating technologies such as OCP-Virt and Hosted Control Planes (HCP aka, "Control Tower").

Product disclaimer

IBM Fusion is being developed and released in an agile manner. In addition to adding new capabilities, the interface is likely to change over time. Therefore, the screenshots used in this document and other lab guides may not always look exactly like what you see in the product. Depending timing, you can expect to encounter some of the following:

- Changes in the user interface (UI), such as the location of buttons or text in various fields
- Additional tabs or buttons

These differences should not affect how the lab exercises work. When changes are observed, have patience and explore.

You will need an IBM account and corresponding IBMid to gain access to IBM Technology Zone.

Getting help

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If you require assistance in interpreting any of the steps in this demo, please post your questions to the #storage_demo_feedback Slack channel for IBMers only. Business Partners can request help at the Partner Plus Support website.

If you have an issue with TechZone or want to open a support case go to the IBM Technology Zone Help. Alternatively, you can send an email to techzone.help@ibm.com. This contact information is also provided in the emails that you receive from IBM Technology Zone.

For help with the Fusion product see the Fusion documentation.

Prerequisites & getting started

You **must** be familiar with the Linux operating system, as well as know how to use the command line interface to execute basic Linux commands such as **kubectl**, **oc**, **exit**, **ls**, **mkdir**, **ssh**, **sudo**, and **curl** (to name few). Ideally, you should also have some experience working with the Red Hat OpenShift Graphical User Interface (GUI) and executing Red Hat OpenShift Container Platform and Kubernetes command line interface (CLI) commands. If you need experience using Linux, Kubernetes and Red Hat OpenShift Container Platform, it is highly recommended that you complete the following Red Hat courses *before* you attempt to work through the exercises in this lab.

Red Hat Courses:

- DO080 Containers Kubernetes and Red Hat OpenShift Technical Overview
- Red Hat OpenShift: Technical Overview

Note Red Hat Partner Connect Training access required.

When instructions are given to enter text, the text is usually shown in mono space font `this is an example`. This is to help indicate that you need to type or copy/paste all the text as is.

IBM Fusion on TechZone

IBM TechZone can be found at this URL https://techzone.ibm.com/. To access IBM TechZone you must have an IBM ID which you can obtain from your IBM representative as a client, Business Partner or employee.

Once you are logged in to IBM TechZone head over to the IBM Fusion Collection and explore the Environement and Resources.

Bookmark these pages for future use or mark them as Favorites by clicking the heart-shaped icon on the

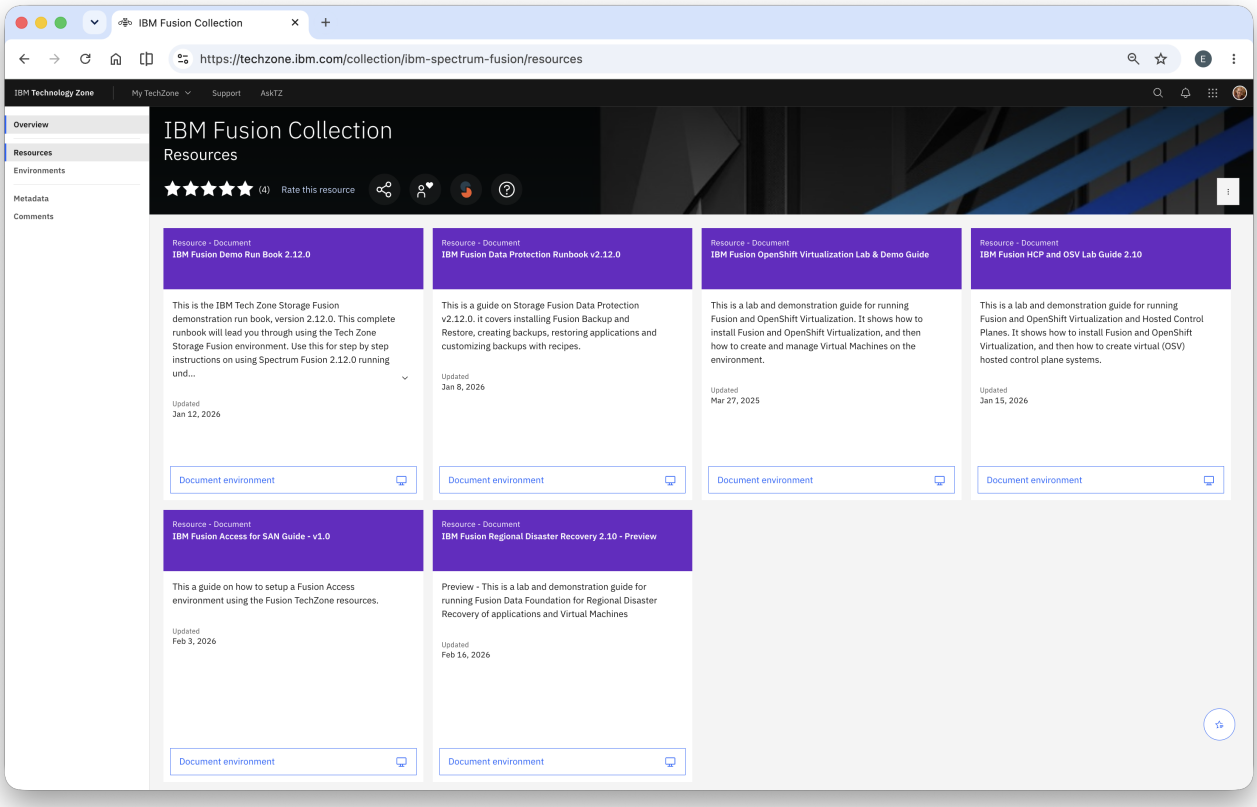
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page.

Resources

The resource page includes links to the available lab guides. These guides are available in PDF format and over time will be also available as on-line viewable material. Due to the rapid development cycles of some products, the guides may not always reflect the latest UI changes and you can expect to encounter some differences. These differences should not affect how the lab exercises work; be patience and explore to enhance your learning.

The lab guides are typically a step by step procedure to performing and learning a set of features and product capabilities. Feel free to explore and experiment!



Environments

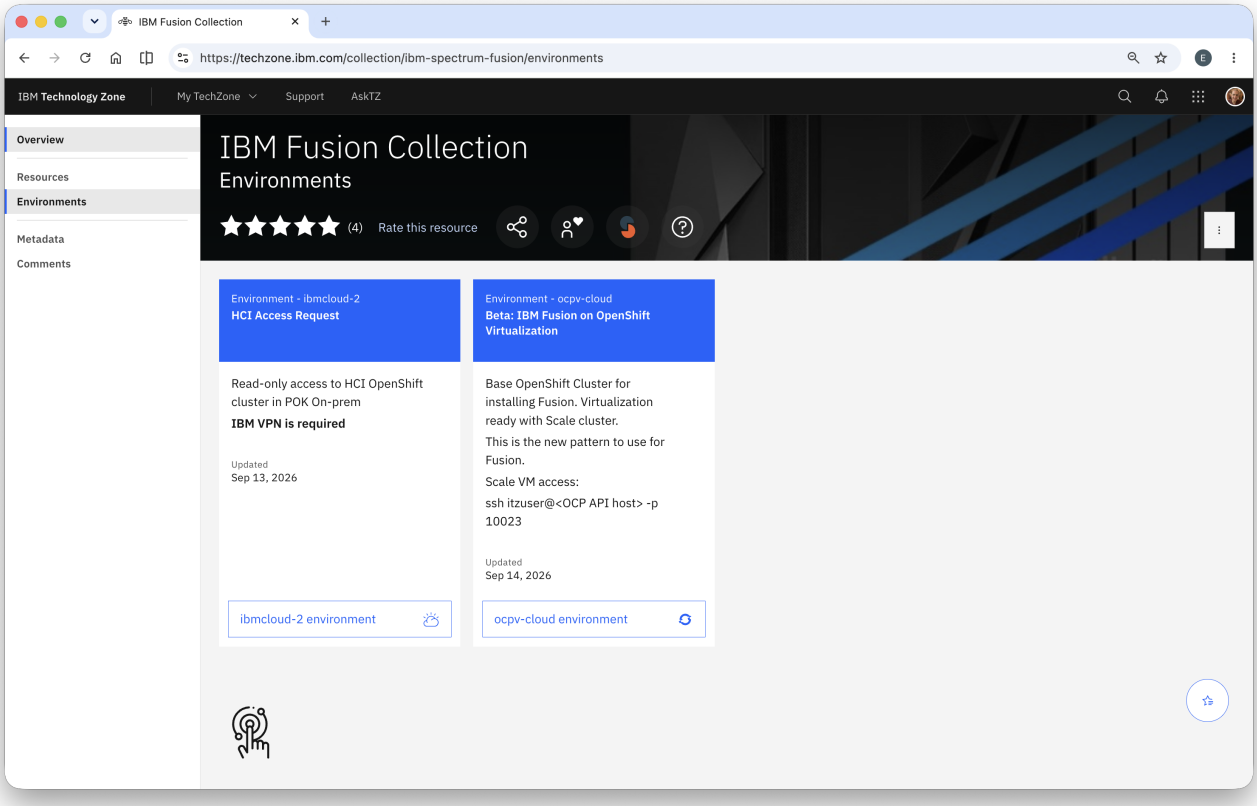
The environments page includes links to available lab resources that you can request for your project. Once your request is provisioned and ready, you will be able to use these resources for your own education, experimentation, and client demos.

The environment that is provisioned for you is an OpenShift (OCP) cluster consisting of your choice of

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components with some extra resources such as a Bastion node and an IBM Storage Scale node. You can use these for the guided labs or use them as a blank slate for your own explorations.

The next sections will walk you through a provisioning and basic use of an OCP cluster on TechZone.



Provisioning a lab

Begin your reservation process by logging in to TechZone and navigating the the Fusion Environments page.

The environments page lists the environment types you can provision and gain access to for your project.

Select the environment titled **Beta: IBM Fusion on OpenShift Virtualization** by clicking Reserve at the bottom of the tile. This will start a provisioning wizard that will walk you through the provisioning process:

1. General

Name your environment with a descriptive name that will help you identify the setup. This name will be visible to all authorized users, including team members, stakeholders, and clients.

Enter a free text description of the environment. Anyone you share this request with will see this

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description when getting access. Include relevant details to help them understand the purpose and expectations.

2. Business purpose

Describe the usage intent, including the context and purpose behind the request. This information helps the TechZone understand the business need and ensure the request is appropriately supported and prioritized.

By choosing a Purpose tile you can indicate the intended use. Each selection has a different reservation time schedule and different requirements for providing an ISC record reference. If you need to provide an ISC record reference simply copy and paste the 18-character Opportunity Number and make sure that the product list in the opportunity include the product name of the lab you are reserving; in our case Fusion.

3. Deployment

TechZone resource reservations rely on IBM Cloud. Choose a location closest to where the session will be hosted (if in person) to optimize performance. While this preference will be considered, deployment to the selected geography is subject to resource availability and cannot be guaranteed.

4. Scheduling

The scheduling page gives you the ability the set the preferred reservation date range within the Policy period determined by your intended use selection you made in the Business purpose page. Provide details on scheduling requirements and the planned dates for the intended request.

Important	Do not finalize dates with clients until the request is approved or until the resources are fully deployed and ready. Always reserve a few days in advance of any meetings or demos.

5. Configuration

Enter clear and detailed configuration choices for the environment request. You can choose to keep the defaults as displayed, or tailor the setup to your needs by clicking Customize and then modifying the environment attributes and then clicking Save.

6. Review and submit

Review the details on this pager before submitting your reservation request. This final step ensures that all required information is accurate and that you are aware of the responsibilities associated with the use of

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IBM TechZone resources.

You can go back and change reservation inputs by clicking the Edit links on the page or the Back button to step backwards in the wizard.

Any wizard step with a blue checkmark icon next to it is properly entered, and any red exclamation point icon should be revisited and corrected before submission.

Once you submit your request the system automatically start a background process to reserve the resources needed for configuring the OpenShift cluster and other resources.

You can track the reservation requests and their status on the My requests page. This page also has a display and sort filter that you can use the manage what you see on the page.

Tip Clear the filter or add the selection of Failed reservations so that you will be able to track reservations that failed to provision properly. This will prevent them from "disappearing" from your view.

You will receive emails from IBM Technology Zone <noreply@techzone.ibm.com> indicating the status of your reservation. The status may indicate a reservation failure or success, and over the duration of use you may also receive reminders for imminent expiration or confirmation of your initiated deletion.

If you receive and expiration notice and you have a date extension option, then follow up and extend your reservation in order to keep using it without disruption.

My reservation failed, what to do?

Reservations may fail from time to time. This depends of many factors including current workloads and available resources. Always reserve in advance of any meetings or demos! if a reservation fails redo the reservation entry using the reservation wizard and described in this section.

Managing a lab

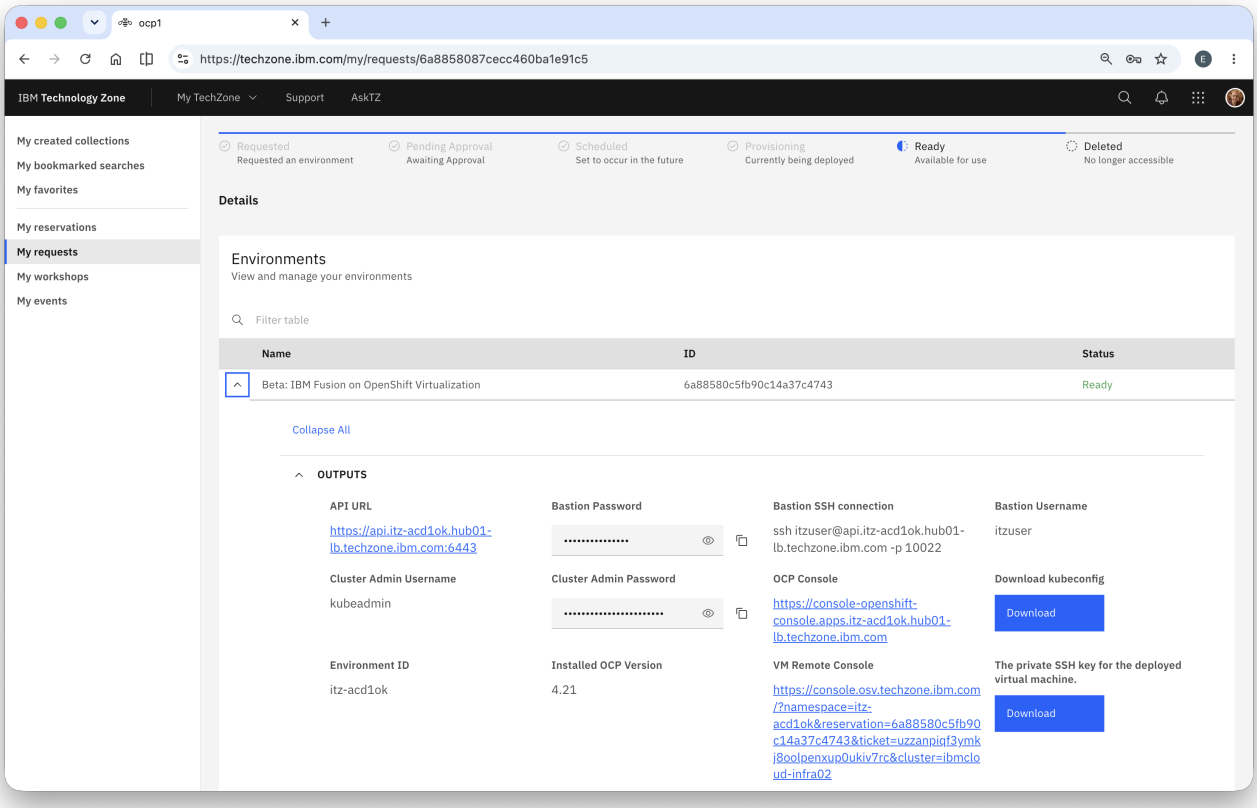
Once an environment reservation is successful you can start using it. If you are a first time then follow the steps below to get started.

Gather up some details

Start on the My requests page, locate the reservation you want to use, and make sure it is in the green "Reservation - Ready" state.

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Open the reservation page by clicking Open this reservation on the reservation tile (or the blue "eye" icon if you are in a list view instead of a tiled view.) Now expand the environment view by clicking the down-pointing arrow next to you environment name.



Before you continue we advise your to collects the various parameters of your environment such as passwords and access key file into a folder on your local desktop system. Create a folder which will be your work folder and collect the following artifacts into it:

- Download the kubeconfig file and save it
- Download the private SSH key and save it
- Create a text file and save the following parameters to it:
  - 24-character environment ID
  - Bastion user `itzuser` and the password from the hidden field
  - OCP cluster user `kubeadmin` and the password from the hidden field

For detailed instructions on connecting to the Web Console, SSH-ing to the Bastion, installing `oc` locally, and VM IP address references, see Accessing the Environment.

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Accessing Your TechZone Lab Environment

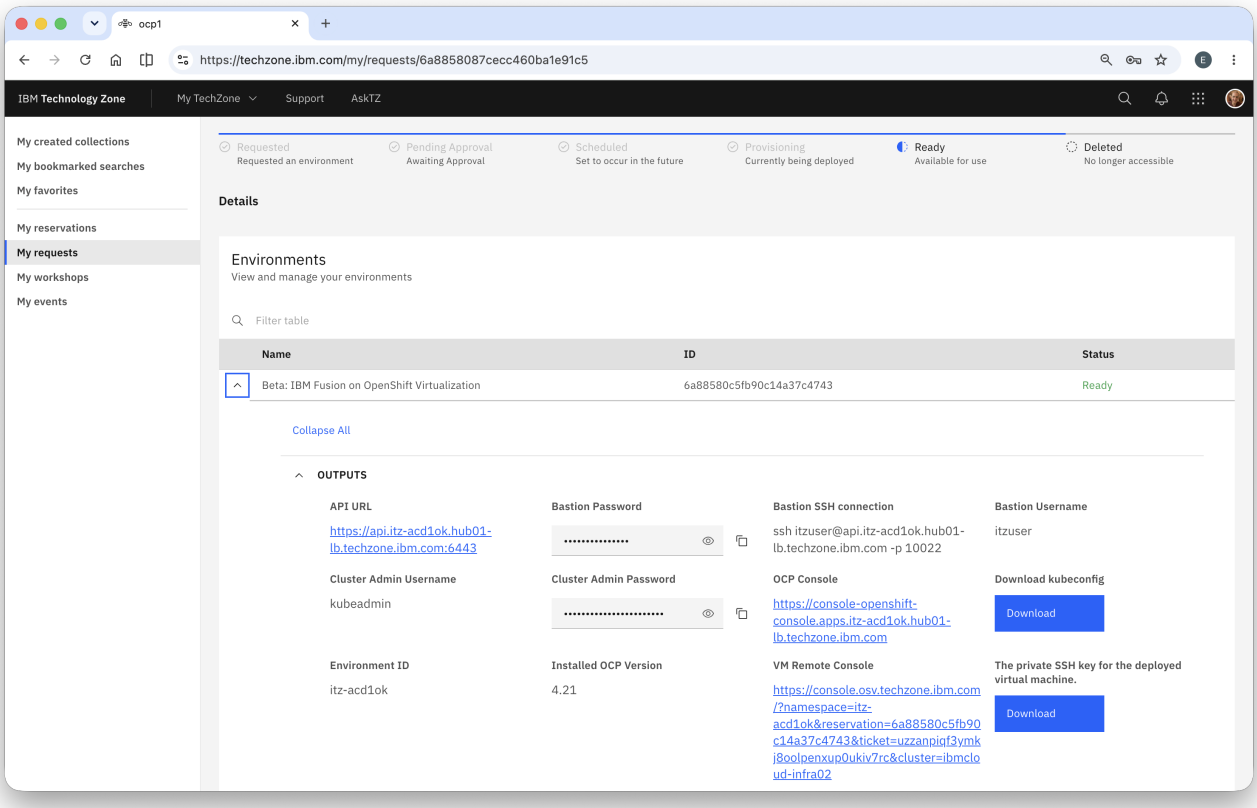
Introduction

Once your TechZone reservation is in the **Ready** state, you have three ways to interact with your OpenShift cluster. Most learners use a combination — the Web Console for UI-driven tasks and either the Bastion or a local `oc` install for CLI work.

Note If you have not yet provisioned a TechZone environment, complete the TechZone Quick Start first.

Gathering Your Credentials

All endpoints and credentials are found in the **Outputs** section of your reservation page. Navigate to My requests, open your reservation, and expand the environment row to reveal the Outputs panel shown below.



Before proceeding, collect the following into a working folder on your local machine:

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- Download the **kubeconfig** file — saves as `conf_kubeconfig_download.conf`
- Download the **private SSH key** — saves as `vm_ssh_key.vm` (check your Downloads folder)
- Note or copy the following values from the hidden fields (click the eye icon to reveal):
  - **Cluster Admin Password** (username is `kubeadmin`)
  - **Bastion Password** (username is `itzuser` — used for `sudo` on the Bastion, not SSH login)
- Note the **Bastion SSH connection** command — it is pre-built with the correct hostname and port
- Note the **API URL** — used for `oc login`

The following table describes every field in the Outputs panel:

Field	Description
API URL	OpenShift API endpoint used for `oc login` and kubeconfig
OCP Console	OpenShift Web Console URL — open directly in your browser
Cluster Admin Username	`kubeadmin`
Cluster Admin Password	Hidden — click the eye or copy icon
Bastion SSH connection	Pre-built SSH command — copy and paste directly into your terminal
Bastion Username	`itzuser`
Bastion Password	Hidden — required only for `sudo` on the Bastion, not for SSH login
Download kubeconfig	Download — saves as `conf_kubeconfig_download.conf`
The private SSH key	Download — required to authenticate SSH to the Bastion
VM Remote Console	Browser-based VNC access to the underlying cluster VMs
Environment ID	Your unique 24-character environment identifier
Installed OCP Version	The OpenShift version running in your cluster

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Important The Bastion host uses **key-based SSH authentication only** on port 10022. Password authentication is disabled on that port. You must download the private SSH key before attempting to connect.

Choosing an Access Method

Method	Best for	Requires
OpenShift Web Console	UI-driven tasks, monitoring, Software Catalog	Browser only
Bastion host (SSH)	CLI tasks — `oc` is pre-installed, no local setup needed	SSH client + private key
Local `oc` on your laptop	CLI tasks without connecting to the Bastion	`oc` installed + kubeconfig

Option 1: OpenShift Web Console

The OpenShift Web Console runs entirely in your browser — no software installation required.

1. From your TechZone reservation page, copy the **OCP Console** URL.
2. Open it in your browser.
3. On the login page, select the **kube:admin** identity provider.

   Tip Select **kube:admin** rather than IBMid to ensure you log in with full `cluster-admin` privileges.

4. Enter `kubeadmin` as the username and paste the **Cluster Admin Password** from your reservation outputs.
5. Upon successful authentication the OpenShift Administrator Dashboard is displayed.

Tip When your browser offers to save the password, accept — you will be logging back in frequently throughout the labs.

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Option 2: Bastion Host via SSH

The Bastion host has `oc`, `kubectl`, `helm`, and `curl` pre-installed. Connecting via SSH is the quickest way to run CLI commands without any local setup.

Step 1: Download the private SSH key

On your TechZone reservation page, click Download next to **The private SSH key for the deployed virtual machine**. The file saves as `vm_ssh_key.vm` (check your Downloads folder).

Step 2: Set key permissions

SSH requires the private key file to be readable only by your user.

macOS / Linux:

```
chmod 400 ~/Downloads/vm_ssh_key.vm
```

Windows (PowerShell):

```
icacls "$env:USERPROFILE\Downloads\vm_ssh_key.vm" /inheritance:r /grant:r
"$env:USERNAME:(R)"
```

Step 3: Connect via SSH

Copy the **Bastion SSH connection** command from your reservation page — it is pre-populated with the correct hostname and port. Add the `-i` flag pointing to your downloaded key:

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10022
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p
10022
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10022.

Step 4: Log in to OpenShift from the Bastion

Once connected to the Bastion, log in to your OpenShift cluster using the **API URL** and **Cluster Admin**

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Password from your reservation:

```
oc login -u kubeadmin -p <cluster-admin-password> https://api.<env-
id>.techzone.ibm.com:6443
```

Verify connectivity:

```
oc whoami
oc get nodes
```

`oc whoami` should return `kubeadmin` and `oc get nodes` should list all cluster nodes in `Ready` state.

Option 3: Local `oc` CLI on Your Laptop

If you prefer to run `oc` commands directly from your laptop without SSH-ing to the Bastion, install the `oc` CLI and use the downloaded kubeconfig.

Step 1: Install the `oc` CLI

macOS:

```
# Using Homebrew
brew install openshift-cli

# Verify
oc version
```

Alternatively, download the binary from the OCP Web Console: click the Help icon (?) in the top-right masthead and select Help › Command Line Tools, then download the macOS binary and move it to `/usr/local/bin/`.

Linux:

```
# Download from the OCP Web Console (Help > Command Line Tools), or use the mirror:
curl -LO https://mirror.openshift.com/pub/openshift-v4/clients/ocp/stable/openshift-
client-linux.tar.gz
tar -xvf openshift-client-linux.tar.gz
sudo mv oc /usr/local/bin/
oc version
```

Windows:

Download the `oc` binary from the OCP Web Console (Help icon → Help › Command Line Tools → **Windows**). Extract the zip and move `oc.exe` to a folder on your `PATH` (e.g. `C:\tools\`).

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Alternatively, install via Chocolatey:

```
choco install openshift-cli
oc version
```

Step 2: Download the kubeconfig

On your TechZone reservation page, click Download next to **Download kubeconfig**. The file saves as `conf_kubeconfig_download.conf`.

Step 3: Connect using the kubeconfig

macOS / Linux:

```
export KUBECONFIG=~/Downloads/conf_kubeconfig_download.conf
oc whoami
oc get nodes
```

Windows (PowerShell):

```
$env:KUBECONFIG = "$env:USERPROFILE\Downloads\conf_kubeconfig_download.conf"
oc whoami
oc get nodes
```

A successful connection returns `system:admin` from `oc whoami`.

Tip

To make the kubeconfig permanent across terminal sessions, merge it into your default `~/.kube/config`:

```
# macOS / Linux
KUBECONFIG=~/.kube/config:~/Downloads/conf_kubeconfig_download.conf \
  oc config view --flatten > /tmp/merged && mv /tmp/merged ~/.kube/config
```

SSH to IBM Storage Scale node

Option 1: Bastion node

To SSH from the Bastion directly into the IBM Storage Scale node:

```
ssh itzuser@10.10.10.202 □
```

□ Verify this IP against the VM Remote Console page before use.

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Option 2: Directly from your workstation

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10023.

VM IP Address Reference

The following IP addresses are assigned to VMs in the standard TechZone Fusion environment. Your configuration may include additional worker nodes — treat this as a representative reference:

VM	IP Address
`itz-*-bastion-1`	`10.10.10.201`
`itz-*-scale-*`	`10.10.10.202`
`itz-*-master-1`	`10.10.10.11`
`itz-*-master-2`	`10.10.10.12`
`itz-*-master-3`	`10.10.10.13`
`itz-*-infra-1`	`10.10.10.2`
`itz-*-infra-2`	`10.10.10.3`
`itz-*-infra-3`	`10.10.10.4`
`itz-*-worker-1`	`10.10.10.21`

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VM	IP Address
`itz-*-worker-2`	`10.10.10.22`
`itz-*-worker-3`	`10.10.10.23`

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IBM Fusion Operator Installation & Console Overview

Introduction

Although your Red Hat OpenShift cluster is provisioned on IBM Technology Zone, IBM Fusion has not yet been deployed. In this guide, you will install the IBM Fusion Operator from the Software Catalog, accept the product license, and explore the core administration interface.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Installing the IBM Fusion Operator

The IBM Operator Catalog and entitlement secrets are pre-configured on your TechZone OpenShift cluster.

Install via OpenShift Web Console

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. In the left navigation menu, go to Ecosystem › Software Catalog.
3. In the search field, type `Fusion` and select the **IBM Storage Fusion** operator tile.
4. Click Install.
5. Leave all installation options as default:
   - **Update channel:** Stable channel (e.g., `v2.x`)
   - **Installation mode:** All namespaces on the cluster (default)
   - **Installed Namespace:** `ibm-spectrum-fusion-ns` (or Operator-recommended namespace)
   - **Approval Strategy:** Automatic
6. Click Install at the bottom of the form.
7. Wait until the Operator status shows **Succeeded** / **Installed operator - ready for use**.

Optional: Install via OpenShift CLI (`oc`)

You can also verify or initiate operator installation from the Bastion terminal:

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```
oc get csv -n ibm-spectrum-fusion-ns
```

Connecting to the IBM Fusion Management Console

IBM Fusion is container-native and integrates directly into the OpenShift console.

1. From the top masthead of the OpenShift Web Console, click the **Application Menu** icon (the 9-square grid).
2. Click **IBM Storage Fusion** from the dropdown menu. A new browser tab opens to the Fusion interface.
3. If prompted with the OpenShift login page, select **kube:admin** and enter your cluster credentials.
4. On first launch, review and select **I have read and accept the license agreement**, then click Continue.

> **Note** Accepting the license initializes the core `SpectrumFusion` Custom Resource (CR) in the `ibm-spectrum-fusion-ns` namespace.

5. You are redirected to the **Welcome to IBM Fusion** dashboard.

Navigating the IBM Fusion Dashboard

The IBM Fusion interface provides a unified view of data management services:

- **Services:** View installed services and install optional capabilities (Fusion Data Foundation, Backup & Restore, Data Cataloging).
- **Storage:** Manage local storage clusters, StorageClasses, and remote IBM Storage Scale file systems.
- **Backup & Restore:** Configure backup locations (S3), retention policies, and application protection.
- **Events:** View Kubernetes-native events with severity filtering (Info, Warning, Critical).
- **Settings:** Review cluster networking, Call Home diagnostics, and cluster interconnect status.
- **Help & Support:** Access the question-mark icon in the masthead to download support logs or link to official IBM documentation.

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Configuring Fusion Data Foundation (FDF)

Introduction

In this guide you will enable the Data Foundation service in the IBM Fusion Console and configure the local Ceph storage cluster using the OpenShift Web Console wizard.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Enabling the Data Foundation Service

Note This guide covers the **Local** configuration mode, the standard mode for single-cluster labs. For an overview of all FDF configuration modes (Local, Dynamic, External, and Provider), see Persistent Storage: Architecture & Concepts.

1. In the IBM Fusion Console, navigate to **Services** in the left navigation menu.
2. Locate the **Data Foundation** service tile, click the tile, and then click Install.
3. On the **Install service** screen, select the **Local** tile and click Install.

   Note Selecting **Local** automatically installs the Local Storage Operator (LSO) — no separate LSO installation step is required.

4. Wait for the service status to show **Healthy**.

Configuring the Ceph Storage Cluster

Once the Data Foundation service is enabled, configure the local Ceph storage cluster through the OpenShift Web Console wizard launched from Fusion:

Step 1: Launch Storage Cluster Creation

1. In the IBM Fusion Console, navigate to Storage › Data Foundation (or **Local storage**).
2. Click Get Started (or Configure Data Foundation / Create Storage Cluster).

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3. This redirects to the **Create StorageSystem** wizard in the OpenShift Web Console.

Step 2: Backing Storage

1. On the **Backing storage** page, select **Create a new StorageClass using local storage devices**.
2. Click Next.

Step 3: Advanced Settings

1. On the **Advanced Settings** page, configure the default StorageClass options:
   - Leave **Use Ceph RBD as the default StorageClass** checked. This sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass after installation.
   - If this cluster will host virtual machines, also check **Set default StorageClass for virtualization**. This marks the RBD virtualization StorageClass as the default for KubeVirt VM disks after installation.
2. Leave all other options at their defaults and click Next.

Note Both default StorageClass options are configured here in the wizard. No additional manual annotation steps are required after installation.

Step 4: Create Local Volume Set

1. Enter `ibm-spectrum-fusion` as the **LocalVolumeSet name**.
2. Select **Disks on selected nodes** and choose the dedicated infra nodes (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`).
3. Set **Disk Type** to **SSD/NVMe**.
4. Click the `^` to reveal the **Advanced** section, and enter "100" in the minimum drive size field; this will exclude the 50GiB drives.
5. Click Next, then click Yes to confirm creation of the LocalVolumeSet.

Step 5: Capacity and Nodes

1. Verify the **Available raw capacity** populated from the discovered disks.
2. Select a **resource profile** — **Balanced** is recommended for lab environments.
3. Click Next.

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Step 6: Security and Network

1. Keep the default encryption settings unless KMS integration (Vault or Thales CipherTrust) is required.
2. Keep the default network option (**Default (OVN)**) unless using Multus.
3. Click Next.

Note If you are configuraing storage for Hosted Control Plane (HCP) then select the "Host" radio button.

Step 7: Data Protection

1. If this cluster will be configured for Regional Disaster Recovery, select **Prepare cluster for disaster recovery (Regional-DR only)**.
2. Otherwise leave unchecked and click Next.

Step 8: Review and Create

1. Review the configuration summary. Click Back to correct any settings.
2. Click Create StorageSystem.

Note Storage cluster initialization takes approximately 10–15 minutes. Wait until the cluster status reports **Ready / Healthy** under Storage › Data Foundation in the OpenShift console before proceeding.

Step 9: Verify StorageClasses and Ceph Cluster Health

From the OpenShift Web Console:

1. Navigate to Storage › StorageClasses and confirm the standard FDF storage classes are listed.
2. Navigate to Storage › Data Foundation and confirm the storage cluster status shows **Ready / Healthy**.
3. Navigate to Ecosystem › Installed Operators › OpenShift Data Foundation and confirm the operator status shows **Succeeded**.

From the CLI:

```
# Check provisioned StorageClasses
oc get sc

# Verify Ceph cluster status
```


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```
oc get storagecluster -n openshift-storage

# Confirm all ODF pods are Running or Completed
oc get pods -n openshift-storage
```

The following standard StorageClasses should now be available:

- `ocs-storagecluster-ceph-rbd` (Block storage for containers & VMs)
- `ocs-storagecluster-cephfs` (Shared file storage / RWX)
- `openshift-storage.noobaa.io` (Object Bucket Claims / S3)

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OpenShift Virtualization with IBM Fusion: Architecture & Overview

Introduction

OpenShift Virtualization (OCP-Virt / OSV) enables enterprises to run and manage legacy virtual machine (VM) workloads alongside modern containerized microservices on a single, unified Red Hat OpenShift platform.

When combined with **IBM Fusion**, VMs benefit from enterprise-grade persistent block storage (Fusion Data Foundation Ceph RBD), rapid VM live migration, snapshotting, and disaster recovery.

Key Capabilities

- **Unified Platform:** Run containers and VMs side-by-side on the same OpenShift cluster, managed through a single control plane.
- **Live Migration:** Move running VMs between worker nodes with zero downtime using KubeVirt's live migration engine — no shared storage or hypervisor-level clustering required.
- **Fusion Storage Integration:** VMs consume Ceph RBD block volumes provisioned by Fusion Data Foundation, providing enterprise snapshotting, cloning, and disaster recovery for VM disks.
- **virtctl CLI:** The `virtctl` command-line tool extends `oc` / `kubectl` with VM-specific operations including console access, start/stop, and live migration.

Architecture

OpenShift Virtualization is built on the **KubeVirt** project. Each VM runs as a standard Kubernetes pod (a `VirtualMachineInstance`) containing a `libvirt` / QEMU process. This means VMs are scheduled, monitored, and managed using the same Kubernetes constructs as containers — namespaces, resource quotas, network policies, and storage classes all apply.

Prerequisites

Before deploying VMs, the following operators must be installed from Ecosystem › Software Catalog and configured on the cluster. Full step-by-step instructions are in Install OCP-Virt.

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Operator	Required	Purpose
IBM Fusion	Yes	IBM Fusion base operator to enable IBM Fusion Console
IBM Fusion Data Foundation	Yes	IBM Fusion Data Foundation data services to provide volumes to virtual machines
IBM Fusion Backup & Restore Hub	Yes	IBM Fusion BnR to provide virtual machine backup and restore capabilities
OpenShift Virtualization	Yes	Provides the KubeVirt engine and the `HyperConverged` cluster resource
MetalLB	Optional	Bare-metal LoadBalancer for exposing VM services externally
Kubernetes NMState	Optional	Node-level network configuration for secondary interfaces (bridges, bonds, VLANs)
Migration Toolkit for Virtualization (MTV)	Optional	Migrates VMs from external sources (OVA/OVF) into OpenShift Virtualization

Important If you have not deployed the first three (3) operators above, please navigate back to the following chapters:

- IBM Fusion Base Operator installation
- IBM Fusion Data Foundation installation
- IBM Fusion Backup & Restore Overview

Software Versions

The following software versions were used to develop and validate the OpenShift Virtualization lab:

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Component	Version
IBM Fusion	2.14.x
Red Hat OpenShift Container Platform	4.21.33
Red Hat OpenShift Virtualization	4.21.17
IBM Fusion	2.14
Migration Toolkit for Virtualization	2.12.9
Kubernetes NMState	4.21.0-202609301335
MetalLB	4.21.0-202609291522

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Installing OpenShift Virtualization

Introduction

In this guide you will install the four operators required to support all OpenShift Virtualization lab exercises, initialize the HyperConverged cluster resource, and set up the shared lab prerequisites (projects, SSH key, `virtctl` CLI).

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Install Required Operators from Software Catalog

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. Navigate to Ecosystem › Software Catalog.
3. Install each of the following operators with default settings (default namespace, automatic approval) by searching for the name and clicking Install:

Operator	Purpose
OpenShift Virtualization	KubeVirt engine and `HyperConverged` cluster resource
Migration Toolkit for Virtualization (MTV)	Migrate VMs from external sources (OVA/OVF) into OSV
MetalLB	Bare-metal LoadBalancer for exposing VM services externally
Kubernetes NMState	Node-level network configuration (bridges, bonds, VLANs)

Initialize the HyperConverged Cluster

1. Navigate to Ecosystem › Installed Operators › OpenShift Virtualization Operator.

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2. Click the **HyperConverged** tab and select Create HyperConverged.
3. Scroll down to the `vmStateStorageClass` field and enter `ocs-storagecluster-cephfs`.

> **Note** `vmStateStorageClass` configures the storage class used to persist the virtual Trusted Platform Module (vTPM) state across VM restarts and live migrations. Required for Windows 2022 VMs.

4. Accept all other defaults and click Create.
5. Wait for the status to show **Available**. When a **Web console update is available** banner appears, click Refresh web console.

Verify from the CLI:

```
oc get hyperconverged -n openshift-cnv
```

The `PHASE` column should show `Available` before proceeding.

Initialize the MTV ForkliftController

1. Navigate to Ecosystem › Installed Operators › Migration Toolkit for Virtualization Operator.
2. Click the **ForkliftController** tab and select Create ForkliftController.
3. Leave the name as `forklift-controller` and click Create.
4. Wait for the status to show **Running**. Refresh the web console when prompted — a new **Migration** menu item will appear in the left navigation.

Verify from the CLI:

```
oc get forkliftcontroller -n openshift-mtv
```

The `STATUS` column should show `Running` before proceeding.

Initialize the NMState Instance

1. Navigate to Ecosystem › Installed Operators › Kubernetes NMState Operator.
2. Click Create instance in the **NMState** tile.
3. Accept all defaults and click Create.
4. Refresh the web console when prompted — **NodeNetworkConfigurationPolicy** and **NodeNetworkState** entries will appear under the **Networking** menu.

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Verify from the CLI:

```
oc get nmstate
```

The `AVAILABLE` column should show `True` before proceeding.

Red Hat OpenShift Virtualization Lab Setup

Create Three Projects

Create three projects that are used across the lab exercises:

```
oc new-project vmlab
oc new-project vmbackup
oc new-project vmnetwork
```

Or from the OpenShift Web Console: navigate to Home › Projects, click Create Project, and create each project in turn.

Create SSH Key for Virtual Machines

You will use an SSH key pair to connect to Linux VMs via `virtctl`. Generate the key pair on the Bastion host:

1. SSH to the Bastion host (see Accessing the Environment).
2. Generate a key pair, accepting all defaults:

   ```
   ssh-keygen -t rsa -b 2048
   ```
3. In the OpenShift Web Console, navigate to Virtualization › Overview › Settings › User.
4. Expand **Manage SSH keys**. Select project `vmlab` and click Not configured under **Authorized SSH key**.
5. Select Add new, then return to your Bastion terminal and print the public key:

   ```
   cat /home/itzuser/.ssh/id_rsa.pub
   ```
6. Copy the output and paste it into the **SSH key** field. Enter `bastion-ssh` as the **Secret Name** and click Save.

Download the `virtctl` CLI

The `virtctl` command-line tool extends `oc` with VM-specific operations such as console access, start/stop, and live migration. Download it directly from your cluster's web console to ensure version

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compatibility with your environment.

1. In the OpenShift Web Console, click the Help icon (?) in the top-right masthead and select **Command Line Tools**.
2. Scroll to the **virtctl – KubeVirt command line interface** section.
3. Right-click the **Download virtctl for Linux for x86_64** link and select **Copy Link Address**.
4. On the Bastion host, download and install `virtctl` using the copied URL:

```
curl -L -O <paste-download-url-here>
tar xzvf virtctl.tar.gz
chmod u+x virtctl
sudo mv virtctl /usr/local/bin/
virtctl version
```


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Deploying, Managing & Migrating Virtual Machines

Introduction

In this lab you will deploy both a Linux and a Windows virtual machine backed by IBM Fusion Data Foundation storage, connect to them using the web console and `virtctl`, and perform a live migration between worker nodes.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note This lab assumes the `vmlab` project and the `bastion-ssh` SSH key secret have been created in Install OCP-Virt. Verify before proceeding.

Create a Linux Virtual Machine

1. Navigate to Virtualization › Catalog.
2. Verify the project is set to `vmlab`.
3. Click the **Template catalog** tab.
4. Check the **RHEL** checkbox under the Operating system filter. Click the **Red Hat Enterprise Linux 9 VM** tile.
5. Enter `rhel9-vmlab` in the **VirtualMachine name** field.
6. Confirm that the `bastion-ssh` SSH key is shown in the configuration summary.
7. Click Quick create VirtualMachine.
8. Wait for the VM status to reach **Running** (approximately 2 minutes).

Verify from the CLI:

Tip Make sure your `KUBECONFIG` environment variable points to $HOME/kubeconfig and you are logged in the Red Hat OpenShift cluster. If not, run the following command:

```
export KUBECONFIG="/home/itzuser/kubeconfig"
```

Verify your virtual machine is running.

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```
oc get vm rhel9-vmlab -n vmlab
```

The `STATUS` column should show `Running` before proceeding.

Connect to the VM

Web Console (VNC / Serial)

1. From the **VirtualMachines** list, click `rhel9-vmlab`.
2. Click the **Console** tab or the **Open web console** link.
3. The **Guest login credentials** panel can be expanded in the upper-left corner. Use the copy buttons to paste the credentials.
4. To switch to a serial console, open the **Console type** drop-down and select **Serial console**.

CLI via `virtctl` SSH

1. On the `rhel9-vmlab` VM details page, click the **Actions** drop-down and select **Copy SSH command (SSH using virtctl)**.
2. SSH to the Bastion host.
3. Paste the command and replace `<path_to_sshkey>` with the actual key path:

```
virtctl -n vmlab ssh cloud-user@vm/rhel9-vmlab \
  --identity-file=/home/itzuser/.ssh/id_rsa
```

4. Enter `yes` when prompted to accept the host fingerprint.

Note If you see `dial tcp [::1]:8080: connect: connection refused`, you are not logged in to the cluster. Run `oc login` on the Bastion first.

Live Migration Between Worker Nodes

Live migration moves a running VM between worker nodes with zero downtime.

Option 1: From the OpenShift Web Console

1. On the `rhel9-vmlab` overview page, note the current **Node** in the General section.
2. Click the **Actions** drop-down and select **Migrate (Migrate to a different Node)**.

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3. Observe the status transition: `Running` → `Migrating` → `Running` on a different node.

Option 2: From the CLI

Apply a `VirtualMachineInstanceMigration` CR to trigger the migration:

```
cat <<'EOF' | oc apply -f -
---
apiVersion: kubevirt.io/v1
kind: VirtualMachineInstanceMigration
metadata:
  name: migration-job
  namespace: vmlab
  labels:
    kubevirt.io/vmi-name: rhel9-vmlab
spec:
  vmiName: rhel9-vmlab
EOF
```

Verify the Migration

From the OpenShift Web Console: On the VM details page, confirm the **Node** field in the General section has changed to a different worker node.

From the CLI: Watch the migration CR until `Phase` shows `Succeeded` using the command below.

```
oc get vm -n vmlab -w
```

Create a Windows Server 2022 Virtual Machine

A pre-built Windows 2022 Server evaluation disk image (qcow2) is used for this exercise.

1. Navigate to Virtualization › Catalog and verify the project is `vmlab`.
2. Click the **Template catalog** tab, then check **Windows** under Operating system filter.
3. Click the **Microsoft Windows Server 2022 VM** tile.
4. Open the **Disk source** drop-down and select **URL (creates PVC)**.
5. Enter the disk image URL in the **Image URL** field:

   ```
   https://ibm.box.com/shared/static/icizg1yn0mx6weeld8xdt91fbesnezbv.qcow2
   ```

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6. Set the virtual machine name field as `win2k22-vmlab`
7. Click Quick create VirtualMachine.
8. Wait for the status to reach **Running** (5–10 minutes — the image downloads and Windows boots).

Verify from the CLI:

```
oc get vm win2k22-vmlab -n vmlab
```

The `STATUS` column should show `Running` before proceeding.

Install Guest Agent & virtio-win Drivers

Windows VMs require the `virtio-win` guest agent to report guest OS details and IP addresses to OpenShift.

1. Click **Open new console** on the VM details page.
2. Use **Send key → Ctrl + Alt + Delete**, then log in with password `Passw0rd` as Administrator.
3. Open **File Explorer → CD Drive (D:) virtio-win** and double-click `virtio-win-guest-tools`.
4. Accept the license, click Install, then Next through all steps, and finish.
5. Return to the VM details page — the **Operating system** and **Hostname** fields will now be populated by the guest agent.

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VM Migration with Migration Toolkit for Virtualization

Introduction

The **Migration Toolkit for Virtualization (MTV)** migrates virtual machines at scale into Red Hat OpenShift Virtualization. In this lab you will simulate migrating a VM from an external source by using an Open Virtual Appliance (OVA) image served over NFS from the Bastion host.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note This lab requires the MTV ForkliftController and the `vmlab` project to be set up. See Install OCP-Virt before proceeding.

Enable NFS on the Bastion Host and Download the OVA

The Bastion host will act as the NFS server that hosts the OVA image for migration.

1. SSH to the Bastion host (see Accessing the Environment).
2. Install the NFS utilities:

```
sudo dnf install nfs-utils.x86_64 -y
```

3. Create the images directory and set permissions so the MTV migration pod can read it:

```
sudo mkdir /images
sudo chmod 2755 /images
sudo chgrp users /images
```

4. Export the `/images` directory via NFS:

```
echo "/images   *(rw,no_root_squash,async)" | sudo tee -a /etc/exports
```

5. Enable the NFS service:

```
sudo systemctl enable --now nfs-server.service
sudo exportfs -a
```

Tip In a real production environment you have to verify the NFS ports are opened through

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the firewall for `nfs`, `mountd` and `rpc-bind`. See List of commands.

6. Verify the export:

```
showmount -e
```

7. Download the OVA image pre-built for this lab:

```
sudo curl -L -o /images/rhel8-vmlab.ova \
  https://ibm.box.com/shared/static/s4ana7xh5a48niz5uf1e66myawbmkqic.ova
```

8. Set read permissions on the OVA so the migration pod can access it:

```
sudo chmod 644 /images/rhel8-vmlab.ova
```

Add the Host Provider

A **Host Provider** represents the destination OpenShift Virtualization cluster.

Note An existing host provider is already present in the `openshift-mtv` namespace. This step creates one scoped to the `vmlab` project for the migration plan.

1. Navigate to Migration › Providers for virtualization.
2. Verify the project is set to `vmlab`. Click Create Provider.
3. Click the **OpenShift Virtualization** tile.
4. Enter `vmlab-host` in the **Provider resource name** field.
5. Select Skip certificate validation.
6. Scroll down and click Create Provider.
7. Wait for the status to transition from **Staging** to **Ready**.

Add the OVA Provider

An **OVA Provider** points to the NFS share containing the OVA image.

1. Navigate to Migration › Providers for virtualization.
2. Verify the project is `vmlab`. Click Create Provider.
3. Click the **Open Virtual Appliance** tile.

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4. Fill in the following fields:

Field	Value
Provider resource name	`bastion-nfs`
URL	`10.10.10.201:/images`

Note `10.10.10.201` is the Bastion host's fixed internal NFS interface in all TechZone OCP-V environments. This address is the same for every reservation and does not need to be changed.

5. Click Create Provider once both fields show green checkmarks.
6. Wait for the provider status to show **Ready** (approximately 2 minutes).
7. Click the **Virtual Machines** tab — the `rhel8-vmlab-4.21-verified` OVA should appear in the list.

Verify both providers are Ready from the CLI:

```
oc get providers -n vmlab
```

Both `vmlab-host` and `bastion-nfs` should show `Ready` in the `STATUS` column before proceeding.

Create and Execute the Migration Plan

1. Navigate to Migration › Plans for virtualization.
2. Verify the project is `vmlab`. Click Create Plan.
3. Enter `rhel8-vmlab-plan` in the **Plan name** field. Feel free to add your own description in the appropriate **Description** field.
4. Select `bastion-nfs` as the **Source provider**.
5. Select `vmlab-host` as the **Target provider**.
6. Select `vmlab` as the **Target project** and click Next.
7. Check the `rhel8-vmlab-4.21-verified` VM checkbox, and click Next.
8. Select **Use new network map** and keep the selected defaults.
9. Enter `networkmap` in the **Network map name** field then click Next.
10. Select **Use new storage map** and make sure the **Target storage** is set to `ocs-storagecluster-`

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`ceph-rbd-virtualization`.

11. Enter `storagemap` in the **Storage map name** field then click Next.
12. Skip the next dialog (`Other settings (optional)`) by clicking Next.
13. Skip the next dialog (`Automation (optional)`) by clicking Next.
14. Skip the next dialog (`Automation (optional)`) by clicking Next.
15. Click Create plan to create the migration plan.
16. On the plan details page, click Start, then confirm with Start in the popup.
17. Click the **Virtual Machines** tab to track the migration pipeline. The migration completes in approximately 5 minutes.

Tip You can also trace the migration execution by monitoring the pod `rhel8-vmlab-plan-[truncated]` and its log within the `vmlab` namespace.

Verify the migration plan status from the CLI:

```
oc get plan rhel8-vmlab-plan -n vmlab -w
```

The `STATUS` column progresses through `Running` to `Succeeded`. You can also watch VM-level progress:

```
oc get vms -n vmlab -w
```

Start and Verify the Migrated VM

1. Navigate to Virtualization › VirtualMachines (project: `vmlab`).
2. Click on `rhel8-vmlab-4.21-verified`. The VM is in a stopped state after migration.
3. Click **Actions → Start**.
4. Once the VM is **Running**, the **Operating system** and **Hostname** fields will populate — this confirms that the `qemu-guest-agent` from the OVA is working.
5. Click **Open web console** and log in with:
   - **Username:** `labuser`
   - **Password:** `Passw0rd`

Firewall Related commands

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To configure your firewall for NFS, use the commands below.

```
sudo firewall-cmd --permanent --add-service=nfs
sudo firewall-cmd --permanent --add-service=mountd
sudo firewall-cmd --permanent --add-service=rpc-bind
sudo firewall-cmd --reload
```

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Virtual Machine Networking

Introduction

OpenShift Virtualization supports multiple networking models for virtual machines — from simple OpenShift Routes for HTTP traffic, to direct external network access via NMState bridge mappings, to bare-metal LoadBalancer services via MetalLB. This lab covers all three.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note This lab uses the `rhel8-vmlab` VM migrated in VM Migration (MTV) and the `vmnetwork` project. Ensure those prerequisites are complete.

Publish an Application with an OpenShift Route

An OpenShift Route exposes an HTTP service running inside the VM to external clients.

1. Navigate to Networking › Services (project: `vmlab`). Click Create Service.
2. In the YAML editor, make the following changes:
   - `metadata.name`: `rhel8-vmlab-http-svc`
   - `spec.selector.app`: `rhel8-vmlab-4.21-verified`
   - `spec.ports[0].targetPort`: `80`
3. Click Create.
4. Click the **Pods** tab to confirm the `virt-launcher` pod for `rhel8-vmlab-4.21-verified` is listed.
5. Navigate to Networking › Routes. Click Create Route.
6. Fill in:
   - **Name:** `rhel8-vmlab-http-svc`
   - **Service:** `rhel8-vmlab-http-svc`
   - **Target Port:** `80 → 80 (TCP)`
7. Click Create.

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Verify from the CLI:

```
oc get svc,route -n vmlab
```

Confirm `rhel8-vmlab-http-svc` appears in both the service and route lists with a populated hostname.

8. Copy the **Location** URL from the Route in the console, or use the hostname from the CLI output above. On the Bastion host, verify the Apache server responds:

   ```
   curl http://<route-location>
   ```

   Expected: a default `index.html` from the Apache HTTP server pre-installed in the OVA.

Tip You can also verify the default webserver page by clicking the URL displayed in **Networking > Routes**.

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Backup & Restore for Virtual Machines

Introduction

IBM Fusion Backup & Restore protects virtual machines in the same way it protects containerized applications — by treating the VM namespace as an application. In this lab you will back up a RHEL 9 VM using a weekly policy and restore it to a new namespace.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note This lab requires the Fusion Backup & Restore service to be installed with a `weekly-backup` policy and `object1` S3 backup location configured. See Locations & Policies before proceeding.

Create and Back Up the VM

1. Switch to the `vmbackup` project using the project selector.
2. Navigate to Virtualization › Catalog › Template catalog. Select the **Red Hat Enterprise Linux 9 VM** template.
3. Enter `rhel9-vmbackup` in the **VirtualMachine name** field and click Quick create VirtualMachine.
4. Wait for the VM to reach **Running** status.
5. Launch the IBM Fusion console from the Application Menu (9-square grid icon in the masthead) → **IBM Storage Fusion**.
6. Navigate to Backup & restore › Applications.
7. Select the checkbox next to `vmbackup` and click Assign backup policy.
8. Select the `weekly-backup` policy checkbox. Leave **Back up now** enabled (green toggle). Click Save.
9. Wait for the backup status to show **Completed** under Backup status.
10. Navigate to Backup & restore › Jobs and click the `vmbackup-weekly-backup-<cluster-name>` job to review the backup summary.

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Restore the VM to a New Namespace

1. Navigate to Backup & restore › Backed up applications and click `vmbackup` to open its details page.
2. Click Restore.
3. Set **Target cluster** to `This cluster`.
4. Select **Create a new project** and enter `vmrestore` in the **Project name** field. Click Next.
5. Select the latest backup time from the list and click Next.
6. Review the **Before restore / After restore** summary and click Restore.
7. Confirm with Restore in the confirmation dialog.
8. Click View job details to monitor progress. Wait for the job to show **Completed**.
9. Return to the OpenShift Web Console. Navigate to Virtualization › VirtualMachines and switch to the `vmrestore` project.
10. Confirm the `rhel9-vmbackup` VM has been restored and is in **Running** status.

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